

ALEX RIZZI

Senior Machine Learning Engineer

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SUMMARY

AI/ML Engineer with 7+ years of experience building and deploying machine learning solutions using Python. Skilled in TensorFlow, PyTorch, and scikit-learn, with expertise in data pipelines, model development, and API integration. Proven leadership in guiding cross-functional teams, mentoring engineers, and driving end-to-end delivery of scalable, high-performance AI systems.

EXPERIENCE

Senior Machine Learning Engineer

MindMatrixSols

 06/2021 - Present

- Mentored engineers and data scientists while collaborating with product teams to align ML initiatives with business KPIs and drive measurable impact.
- Developed an LLM-based automation system for RFP generation. Reduced proposal drafting time by 70% and improved content quality and compliance using structured prompt chaining.
- Built and deployed a Text-to-SQL chatbot using an open-source Hugging Face model that translates natural English into complex SQL queries, enabling non-technical users to perform data analysis and gain insights independently.
- Built an automation system for medical claims using Agentic AI. A RAG chatbot validates submissions, OCR reads documents, and a search tool verifies medication details.
- Reducing model training time by approximately 40%.
- Built robust MLOps pipelines using Kubernetes, MLflow, and Airflow, improving deployment reliability and minimizing model drift.
- Streamlined ETL processes and data lakes on AWS (Redshift, S3, Glue), enhancing data availability for modeling.
- Used Databricks with MLflow for model tracking and experiment management. Ensured reproducibility, version control, and team collaboration.

Data Scientist & AI/ML Specialist

Reborn Code

 02/2019 - 05/2021

- Engineered and deployed end-to-end machine learning solutions using Python. TensorFlow, PyTorch, scikit-learn, and Pandas for applications like demand forecasting and customer segmentation.
- Built automated training and evaluation pipelines with NumPy, Pandas, and MLflow, reducing model retraining time by approximately 25% and improving reproducibility.
- Engineered an end-to-end sales forecasting pipeline on Microsoft Fabric for 1 million+ customers, slashing the Mean Absolute Percentage Error (MAPE) from 46% to under 15% for the top customer segment responsible for 75% of business revenue.
- Built an end-to-end sales forecasting pipeline on Microsoft Fabric, reducing MAPE from 46% to under 15% for the top 75% revenue segment.
- Enhanced model performance through advanced feature engineering, hyperparameter tuning, cross-validation, and data preprocessing in Python.
- Applied MLOps best practices including model versioning, CI/CD pipelines, and monitoring with MLflow, reducing drift and deployment errors.

Machine Learning Engineer

Xcira

 02/2017 - 01/2019

- Developed scalable ML testing and validation frameworks using Python, Pytest, NumPy, Pandas, improving model reliability and evaluation coverage.
- Automated testing of 100 ML workflows and pipelines built with scikit-learn, TensorFlow, and PyTorch, enhancing model stability and deployment readiness by over 15%.
- Integrated continuous testing of ML pipelines with Azure DevOps, Docker, and MLflow, optimizing workflow automation and deployment reliability.
- Designed and executed data validation and ETL testing strategies in MySQL, PostgreSQL, and Pandas, ensuring accurate inputs and robust backend support for ML models.
- Built time series, price elasticity, and recommendation models with Python, PySpark, and deep learning to improve accuracy and business outcomes.
- Developed and deployed deep learning and computer vision models for image/text analysis using AWS SageMaker and GCP Vertex AI.
- Built HR automation bots and ML pipelines with MLOps and feature engineering to improve scalability and efficiency.

SKILLS

Programming & Scripting

Python	JavaScript	TypeScript	SQL
Bash			

Machine Learning & Data Science

TensorFlow	Deep Learning	PyTorch	
Scikit-learn	Pandas	NumPy	SciPy
OpenCV	Matplotlib	Seaborn	
Machine Learning	LlamaIndex	NLP	
LangChain	OpenAI		
Prompt Engineering	Crew AI	Streamlit	
Databricks			

Cloud & MLOps

AWS EC2	Lambda	SageMaker	
GCP Vertex AI	BigQuery	Azure ML	
Docker	Kubernetes	MLflow	
Kubeflow	Airflow	CI/CD	GraphQL
ETL Pipelines	OAuth/JWT	FastAPI	
Flask	Django	RESTful API Design	

Databases & Storage

AWS EC2	S3	RDS	CodePipeline
GCP	Docker	Kubernetes	Terraform
Jenkins	GitLab CI	Travis CI	Heroku
Monitoring	Datadog		

Data Science

Time Series Forecasting	PySpark
Price Elasticity Modeling	Computer Vision
HR Automation Bots	

EDUCATION

Bachelor's Degree